

Computational Methods in Business Analytics

BT3102

About this Course

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Tutorial:

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Office Hours: Wed: 16:00-17:00

Email us at any time!

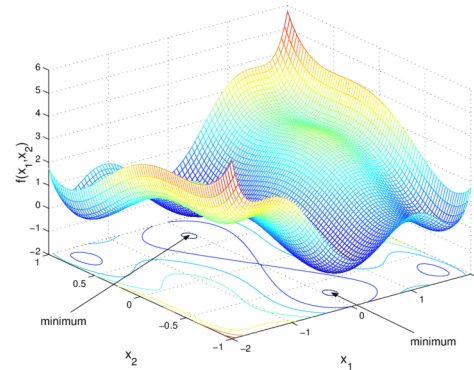
Class participation + in-class quiz: 5%,

4 homework assignments 45%

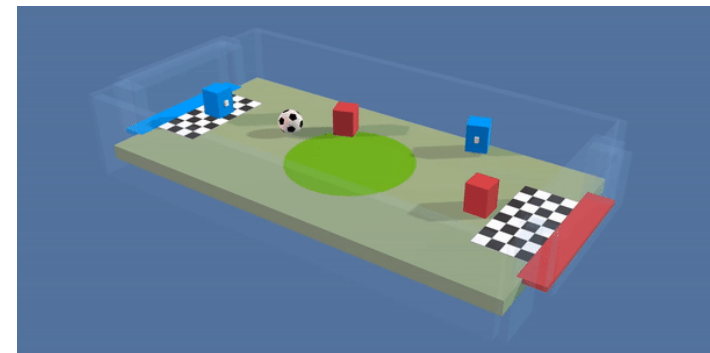
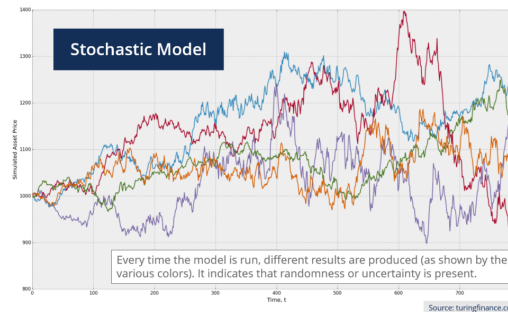
Final Exam 50%

Topics to Cover

- Nonlinear Optimization
- Randomized Experiments
- Stochastic Models and Simulation
- Markovian Decision Processes and Dynamic Programming



SEGMENTING YOUR
A/B TEST RESULTS



Mathematics Refresh: Multivariable Calculus

The (negative) gradient of a function is its steepest (descent) ascent direction.

The (negative) gradient of a function is perpendicular to the level curve.

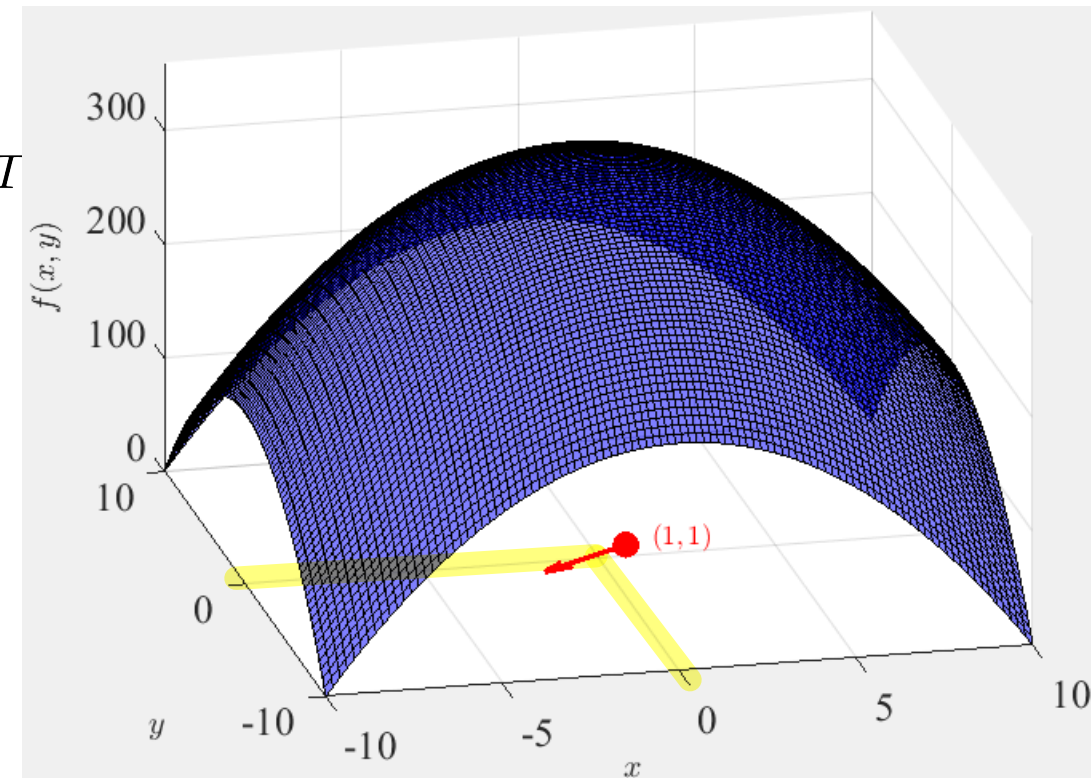
Given a differentiable function $f(\mathbf{x}) : \mathbb{R}^n \rightarrow \mathbb{R}$, the gradient is $\nabla f = \left[\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_n} \right]^T$

e.g., $f(x, y) = 350 - 2x^2 - 1.5y^2$

$$\Rightarrow \nabla f(x, y) = [-4x, -3y]^T$$

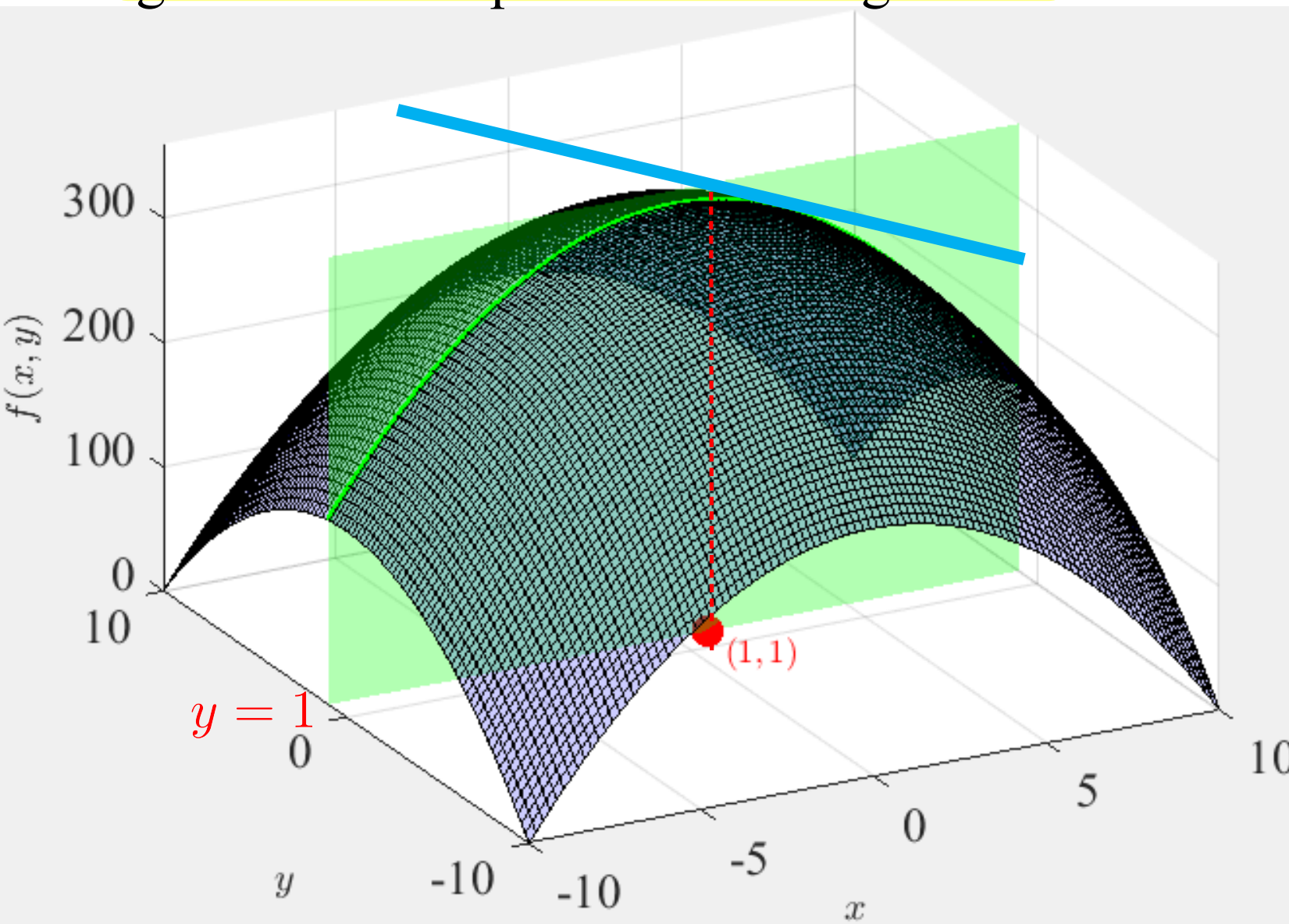
e.g., $(x_0, y_0) = (1, 1) \Rightarrow \nabla f(1, 1) = [-4, -3]^T$
any point on x y plane

Why does following the direction of the gradient at a point give the steepest increase of the function?



Mathematics Refresh: Multivariable Calculus

The geometric interpretation of the gradient:



The rate of change of the objective function $f(x, y)$ when you move slightly along the coordinate axes $y=1$, move along x

$$f(x, y) = 350 - 2x^2 - 1.5y^2$$

$$\Rightarrow \nabla f(x, y) = [-4x, -3y]^T$$

$$(x_0, y_0) = (1, 1)$$

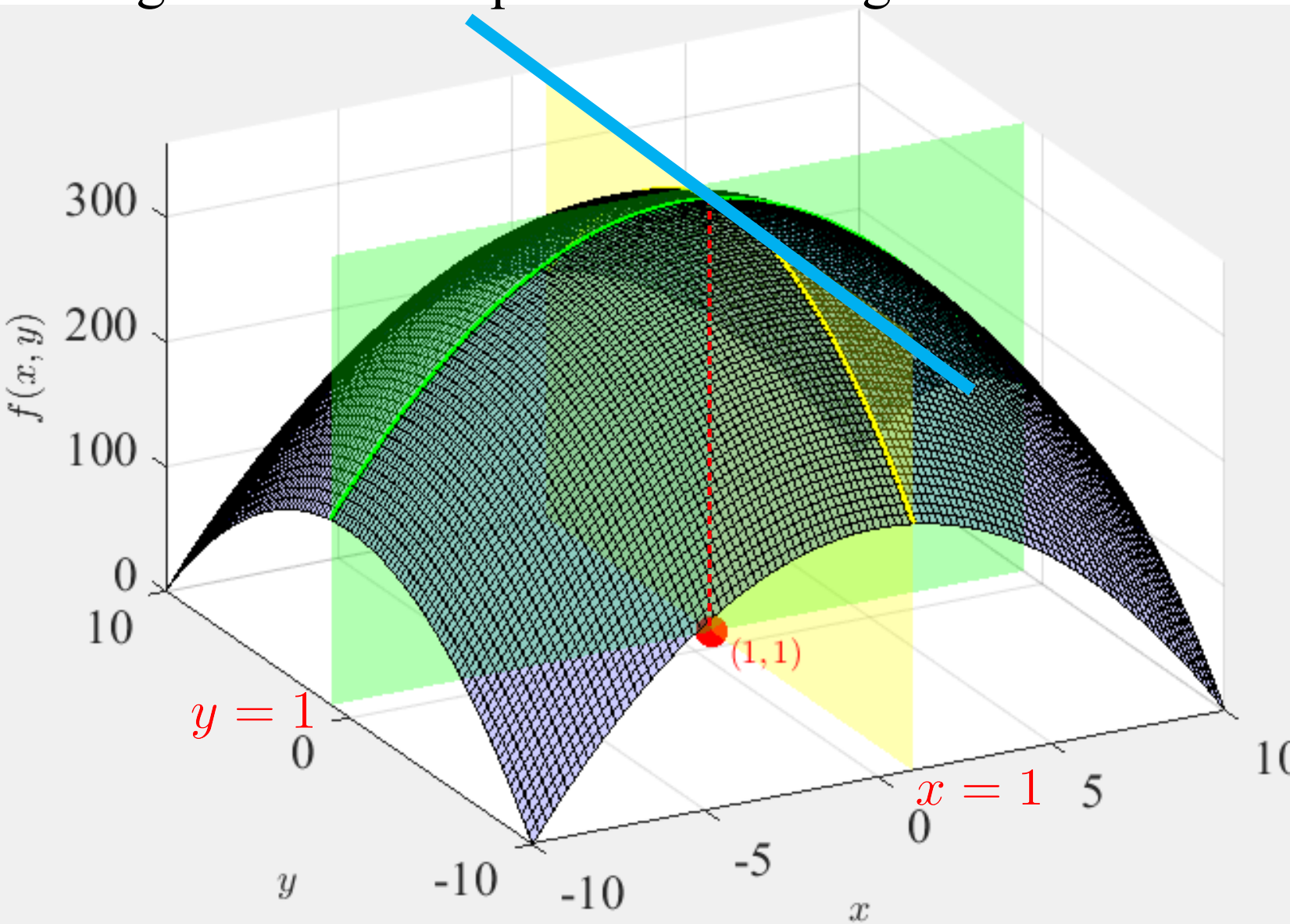
$$\Rightarrow \nabla f(1, 1) = [-4, -3]^T$$

The slope of this green curve at $x=1$ is:

$$\left. \frac{\partial f(x, y)}{\partial x} \right|_{x=1} = -4x \Big|_{x=1} = -4$$

Mathematics Refresh: Multivariable Calculus

The geometric interpretation of the gradient:



The rate of change of the objective function when you move slightly along the coordinate axes

$f(x, y)$
 $x=1$, move along y

$$f(x, y) = 350 - 2x^2 - 1.5y^2$$

$$\Rightarrow \nabla f(x, y) = [-4x, -3y]^T$$

$$(x_0, y_0) = (1, 1)$$

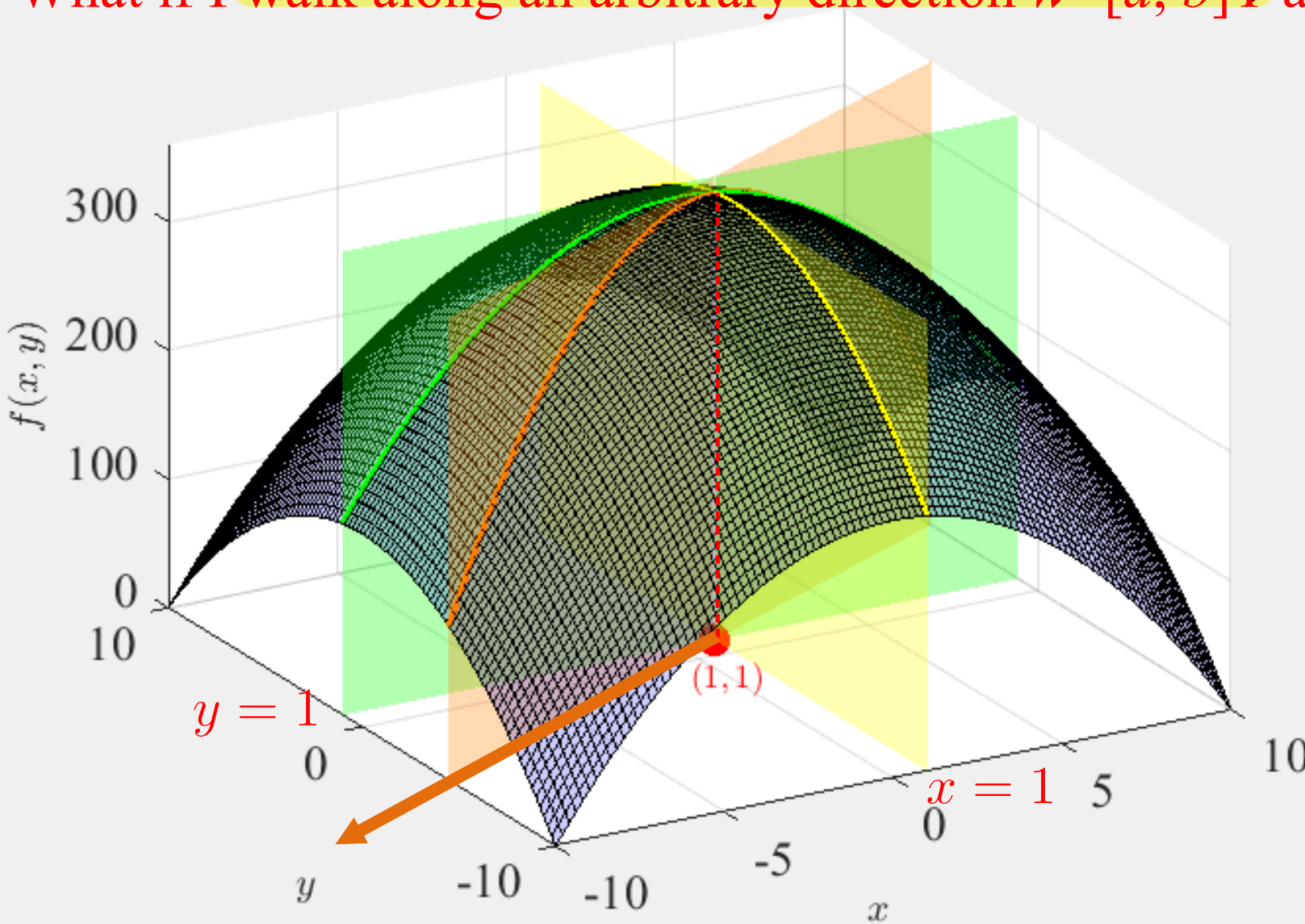
$$\Rightarrow \nabla f(1, 1) = [-4, -3]^T$$

The slope of this yellow curve at $y=1$ is:

$$\left. \frac{\partial f(x, y)}{\partial y} \right|_{y=1} = -3y \Big|_{y=1} = -3$$

Mathematics Refresh: Multivariable Calculus

What if I walk along an arbitrary direction $\mathbf{w} = [a, b]^T$ at my current point for a small step size, then what should be the rate of change of the objective function?



$$f(x, y) = 350 - 2x^2 - 1.5y^2$$

$$\Rightarrow \nabla f(x, y) = [-4x, -3y]^T$$

$$(x_0, y_0) = (1, 1)$$

$$\Rightarrow \nabla f(1, 1) = [-4, -3]^T$$

Mathematics Refresh: Multivariable Calculus

Directional Derivative

Recall the definition of partial derivative: current point (x_0, y_0)

$$\left. \frac{\partial f(x, y)}{\partial x} \right|_{(x_0, y_0)} = \lim_{h \rightarrow 0} \frac{f(x_0 + \overset{\text{small step}}{ah}, y_0) - f(x_0, y_0)}{\underset{\text{step size}}{h}} \quad \text{rate of change of the fn along } x\text{-axis while holding } y\text{-value fixed} \quad \text{direction: } \mathbf{u} = [a, 0]^T$$

$$\left. \frac{\partial f(x, y)}{\partial y} \right|_{(x_0, y_0)} = \lim_{h \rightarrow 0} \frac{f(x_0, y_0 + bh) - f(x_0, y_0)}{h} \quad \text{direction: } \mathbf{v} = [0, b]^T$$

$$\underset{\substack{\text{Directional Derivative} \\ \text{of } f(x, y)}}{\nabla_{\mathbf{w}} f(x_0, y_0)} = \lim_{h \rightarrow 0} \frac{f(x_0 + ah, y_0 + bh) - f(x_0, y_0)}{h} \quad \text{direction: } \mathbf{w} = [a, b]^T$$

Hard to compute

Mathematics Refresh: Multivariable Calculus

Directional Derivative

$$\begin{aligned}\nabla_{\mathbf{w}} f(x_0, y_0) &= \lim_{h \rightarrow 0} \frac{f(x_0 + ah, y_0 + bh) - f(x_0, y_0)}{h} \quad \text{direction: } \mathbf{w} = [a, b]^T \\&= \lim_{h \rightarrow 0} \frac{g(h) - g(0)}{h} \\&= g'(0) \quad \text{derivative of function } g \text{ at point } 0\end{aligned}$$

Define $g(h) = f(x_0 + ah, y_0 + bh)$
 $\Rightarrow g(0) = f(x_0, y_0)$
 $x \cdot \frac{dx}{dh} = a$ $y \cdot \frac{dy}{dh} = b$

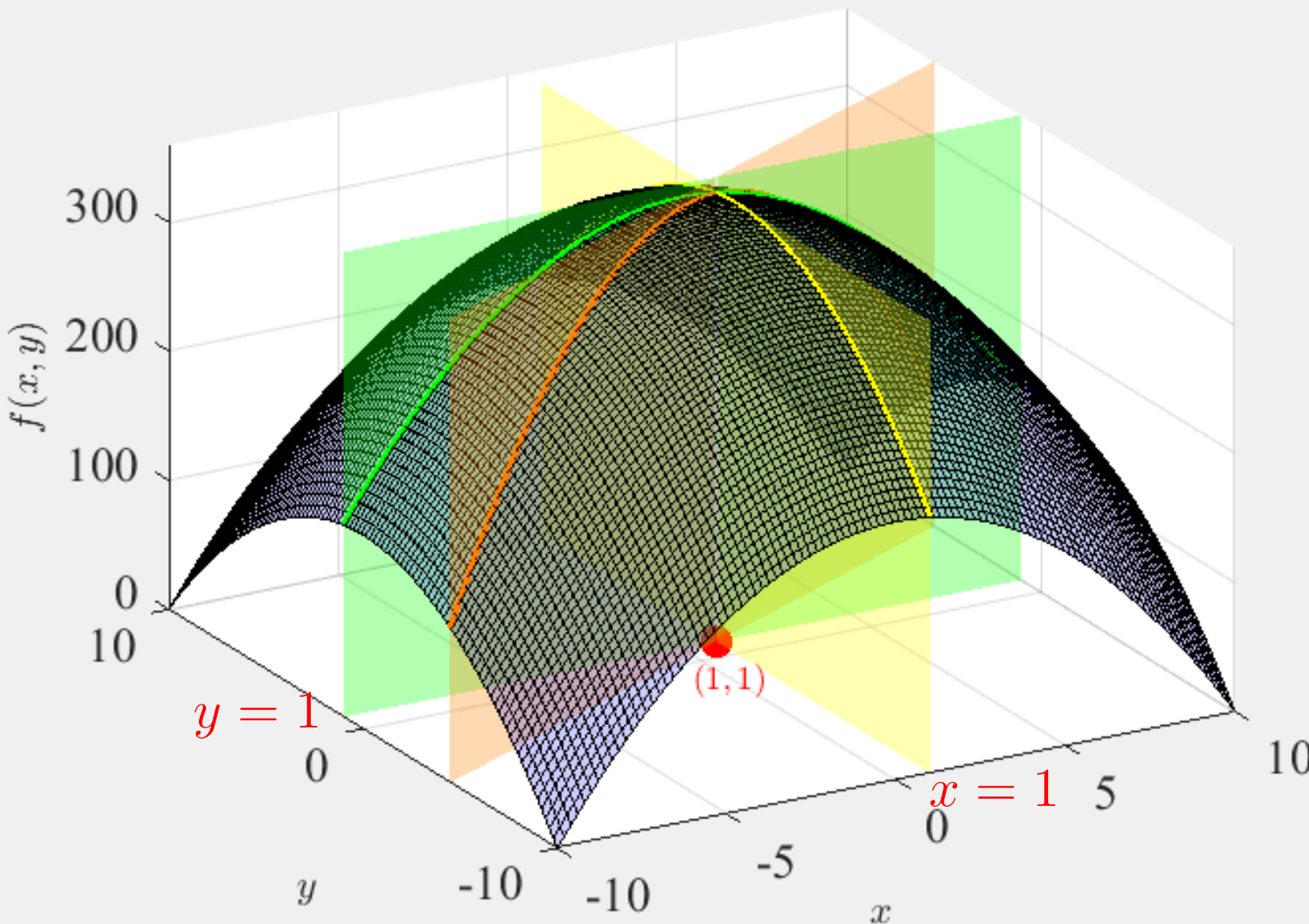
recall: $f'(x) = (g(h(x)))' = g'(h(x)) h'(x)$
- keep the inside multiply by derivative of the inside
- take derivative of outside

$$\begin{aligned}g'(h) &= \frac{dg(h)}{dh} = \frac{\partial f(x, y)}{\partial x} \frac{dx}{dh} + \frac{\partial f(x, y)}{\partial y} \frac{dy}{dh} \quad \text{by chain rule} \\&= \frac{\partial f(x, y)}{\partial x} a + \frac{\partial f(x, y)}{\partial y} b \\ \therefore g'(0) &= \left. \frac{\partial f(x, y)}{\partial x} \right|_{(x_0, y_0)} a + \left. \frac{\partial f(x, y)}{\partial y} \right|_{(x_0, y_0)} b = f_x(x_0, y_0) a + f_y(x_0, y_0) b \\&= [f_x(x_0, y_0), f_y(x_0, y_0)] \cdot \mathbf{w} = \nabla f(x_0, y_0) \cdot \mathbf{w}\end{aligned}$$

h=0, x=x0, y=y0

Mathematics Refresh: Multivariable Calculus

Directional Derivative $\nabla_{\mathbf{w}} f(x_0, y_0) = \nabla f(x_0, y_0) \cdot \mathbf{w}$



$$f(x, y) = 350 - 2x^2 - 1.5y^2$$

$$\Rightarrow \nabla f(x, y) = [-4x, -3y]^T$$

$$(x_0, y_0) = (1, 1)$$

$$\Rightarrow \nabla f(1, 1) = [-4, -3]^T$$

$$\mathbf{w} = [-5, -2]^T$$

$$\Rightarrow \nabla_{\mathbf{w}} f(x_0, y_0)$$

$$= \nabla f(x_0, y_0) \cdot \mathbf{w}$$

$$= [-4, -3] \cdot [-5, -2] = 26$$

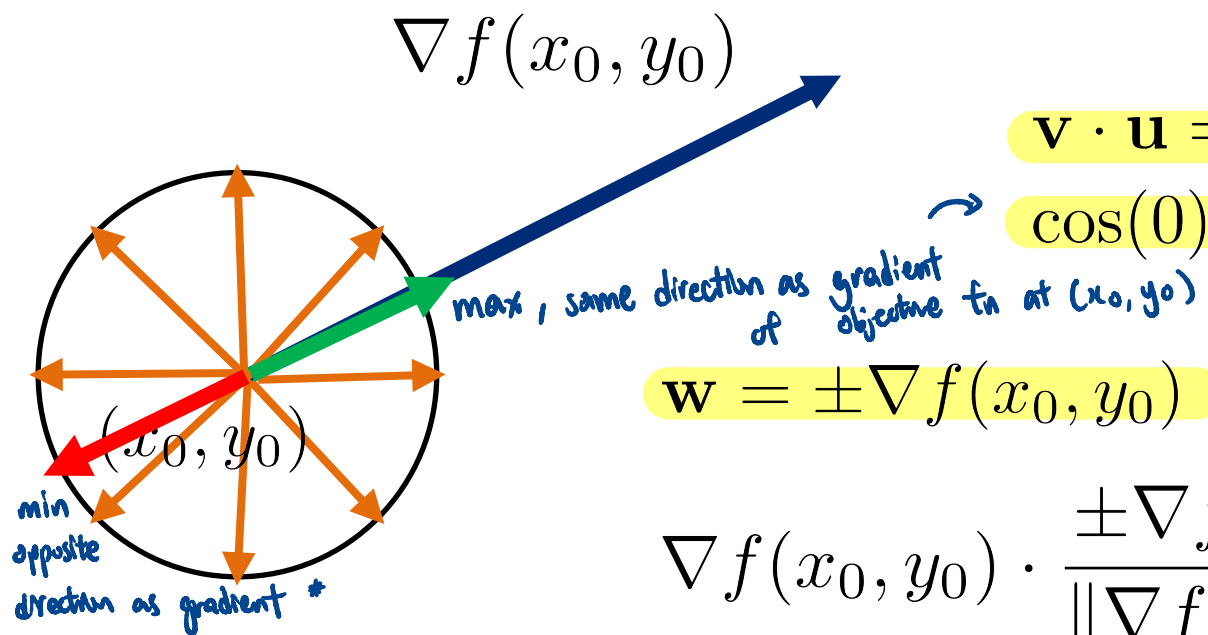
Mathematics Refresh: Multivariable Calculus

Directional Derivative $\nabla_{\mathbf{w}} f(x_0, y_0) = \nabla f(x_0, y_0) \cdot \mathbf{w}$

The (negative) **gradient** of a function is its **steepest** (descent) ascent direction.

direction.

$$\max_{\mathbf{w}} \left(\text{or } \min_{\mathbf{w}} \right) \nabla f(x_0, y_0) \cdot \left(\frac{\mathbf{w}}{\|\mathbf{w}\|} \right) \quad \begin{array}{l} \text{unit vector} \\ \text{A vector of} \\ \text{length 1} \end{array} \quad \begin{array}{l} \mathbf{w} = [-5, -2]^T \\ \frac{\mathbf{w}}{\|\mathbf{w}\|} = [-5/\sqrt{29}, -2/\sqrt{29}]^T \end{array}$$



$$\mathbf{w} = \pm \nabla f(x_0, y_0)$$

$$\nabla f(x_0, y_0) \cdot \frac{\pm \nabla f(x_0, y_0)}{\|\nabla f(x_0, y_0)\|} = \pm \|\nabla f(x_0, y_0)\|$$

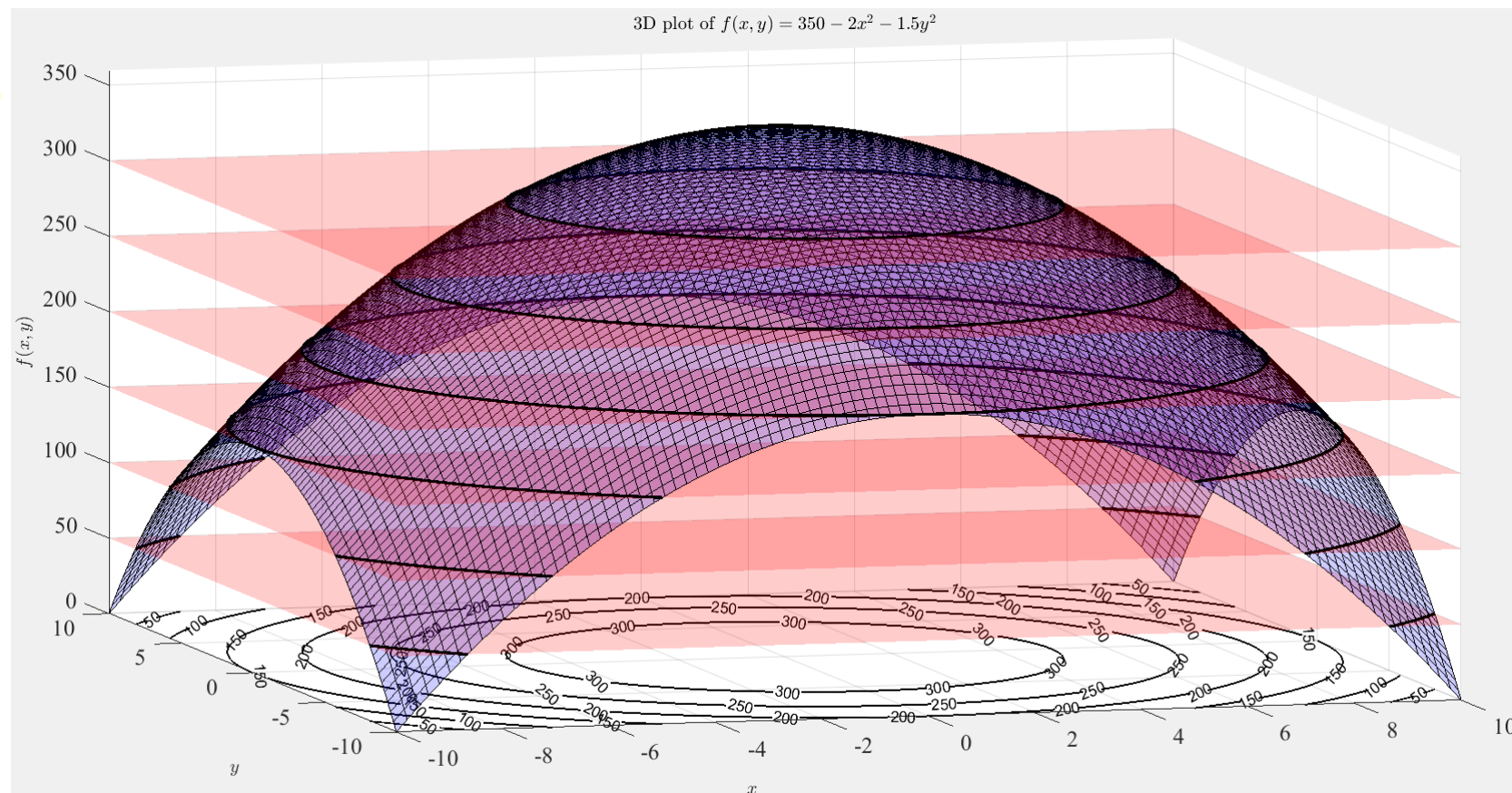
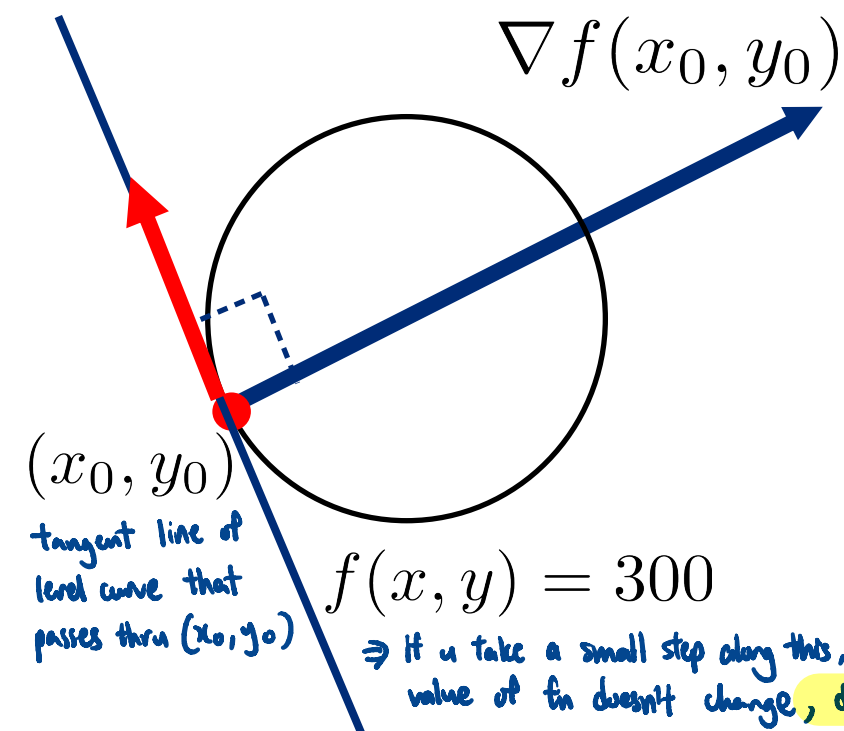
Mathematics Refresh: Multivariable Calculus

Directional Derivative $\nabla_{\mathbf{w}} f(x_0, y_0) = \nabla f(x_0, y_0) \cdot \mathbf{w}$

The (negative) **gradient** of a function is **perpendicular to the level curve**.

$$\nabla f \cdot \mathbf{u} = \|\nabla f\| \cdot \|\mathbf{u}\| \cos \theta$$

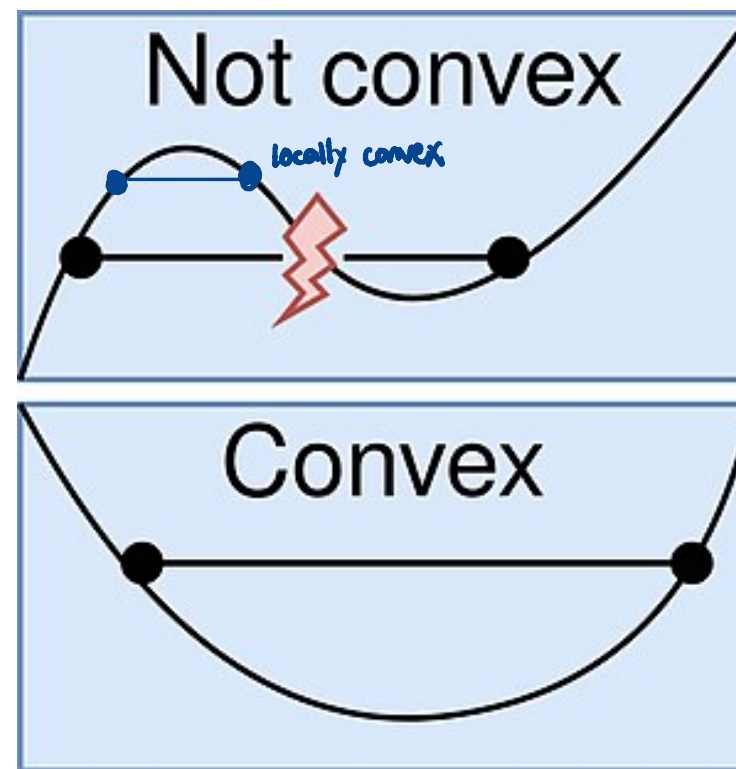
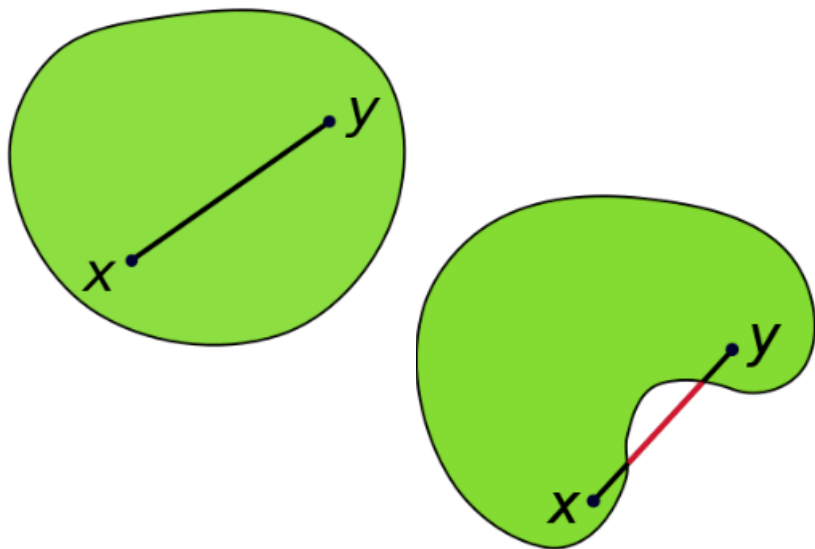
$$\cos(\pi/2) = 0 \quad *$$



* Level curve $f(x, y) = c$

Mathematics Refresh: Multivariable Calculus

Convex Set: S is a convex set if for all x and y in S , the line segment connecting x and y is included in S .



Convex Function: $f : S \rightarrow \mathbb{R}$, where S is a convex set and f satisfies:
 $\forall t \in [0, 1], \forall x_1, x_2 \in S : f(tx_1 + (1 - t)x_2) \leq tf(x_1) + (1 - t)f(x_2)$

Mathematics Refresh: Multivariable Calculus

$C1$ function: 1st-order derivative is a continuous function

$C2$ function: 2nd-order derivative is a continuous function

Taylor's Theorem $\Rightarrow$ express an ugly general function as the sum of a bunch of polynomials

1st-order approximation

$$f(\mathbf{w}) \approx f(\mathbf{w}_0) + \nabla f^T(\mathbf{w}_0)(\mathbf{w} - \mathbf{w}_0)$$

2nd-order approximation

$$f(\mathbf{w}) \approx f(\mathbf{w}_0) + \nabla f^T(\mathbf{w}_0)(\mathbf{w} - \mathbf{w}_0) + \frac{1}{2}(\mathbf{w} - \mathbf{w}_0)^T \nabla^2 f(\mathbf{w}_0)(\mathbf{w} - \mathbf{w}_0)$$

more terms $\rightarrow$ more accurate approx.

Mathematics Refresh: Linear Algebra

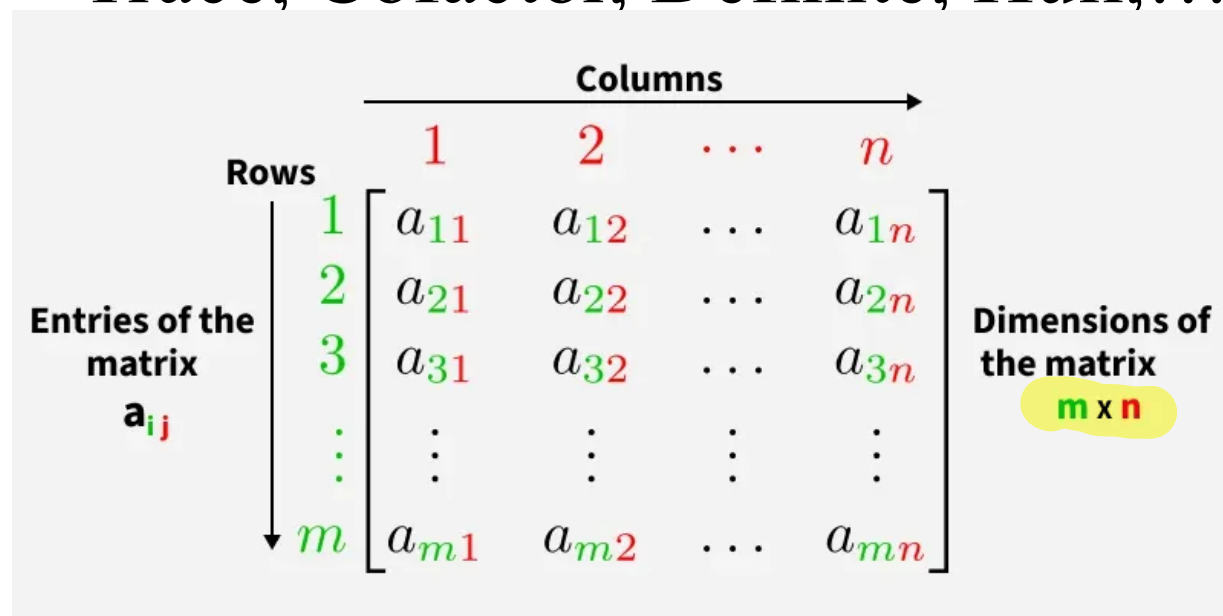
2nd-order approximation

$$f(\boldsymbol{w}) \approx f(\boldsymbol{w}_0) + \nabla f^T(\boldsymbol{w}_0)(\boldsymbol{w} - \boldsymbol{w}_0) + \frac{1}{2}(\boldsymbol{w} - \boldsymbol{w}_0)^T \overset{\text{Hessian}}{\nabla^2} f(\boldsymbol{w}_0)(\boldsymbol{w} - \boldsymbol{w}_0)$$

Matrix and Linear Algebra

1	0	4	2
1	2	6	2
2	0	8	8
2	1	9	4

Number of Inversions, Rank,
Trace, Cofactor, Definite, Hull,...

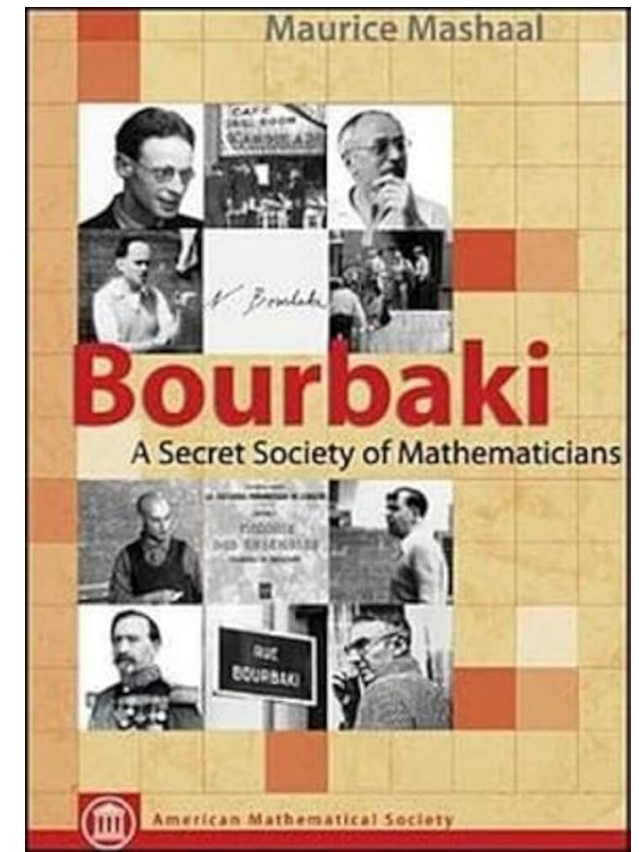


What, exactly, is a matrix?

Why is matrix multiplication defined in precisely this peculiar way?

Why the matrix $P^{-1}AP$ is “similar” to A ?
What exactly does similar mean here?

What is the true nature of eigenvalues and eigenvectors? $Ax = \lambda x$



They think intuition and abstraction as mortal enemies. And when forced to choose between the two, they sacrifice intuition

Space: start with a set, which is a collection of objects. Then you equip that set with some additional structure

e.g., topological space, vector space, norm space, etc.

“Space” encourages us to think geometrically

3D Euclidean Space:

- (a) we can define distances and angles.
- (b) objects can be moved or transformed from one position to another.

Intuition (not maths definition): supporting motion is the core feature of a space.

Vector Space (Linear Space)

1. What kind of objects live in a vector space? What do these objects have in common?

2. How do objects move inside a vector space? What is the correct language for describing those movements?

Vectors describe objects; Matrices describe linear transformations

1. What kind of objects live in a vector space? What do these objects have in common?

Let $B = (b_1, b_2, \dots, b_n)$ be a basis, then every object v in the space can be written uniquely as $v = x_1 b_1 + x_2 b_2 + \dots + x_n b_n$
coords of object v written in language of B

$$[v]_B = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

For a finite-dimensional vector space: After choosing a basis, every object can be represented by a vector.

Different objects become different columns of numbers.

Different bases give different coordinate descriptions.

And the order of the coordinates carries the meaning.

2. How do objects move inside a vector space? What is the correct language for describing those movements?

Linear Transformation: Ax

Vectors describe objects. Matrices describe transformations. Matrix-vector multiplication performs them

What makes a transformation linear? *it respects the linear combination*

$$T(ax + by) = aT(x) + bT(y)$$

Mathematics Refresh: Linear Algebra

A matrix is *one* description of a linear transformation under a chosen coordinate system



Choose one basis and a linear transformation is described by one matrix. Choose another basis and the same transformation is described by another matrix

Mathematics Refresh: Linear Algebra

If A and B are two matrices describing the same linear transformation, obtained by choosing two different bases, then there exists an invertible matrix P such that $A = P^{-1}BP$

A

$\Rightarrow$ similar matrices

Matrix P is interpreter standing between matrices A and B that adjusts the 2 coordinate systems

$\hookrightarrow$ differently coordinated descriptions of the same linear transf.

Given a basis, a matrix can describe a linear transformation. But a matrix can also describe a basis $\Rightarrow$ if the cols. of an invertible matrix are linearly independent

The same matrix can be read in two different ways:
as the description of a transformation;
or as the description of a coordinate system

Mathematics Refresh: Linear Algebra

$$a = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} \quad M = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} = \begin{bmatrix} | & | & & | \\ m_1 & m_2 & \cdots & m_n \\ | & | & & | \end{bmatrix}$$

If the column vectors of M are linearly independent, then they form a basis. A basis is, in effect, a coordinate system. Each basis vector defines one coordinate direction.

Moving the object is equivalent to moving the coordinate system (motion is relative).

Mathematics Refresh: Linear Algebra

Suppose we want the coordinate numbers of a point to change from $a=(1,1)$ to $b=(2,3)$

Method 1: keep the coordinate system fixed and move the point, apply transformation $M = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$

$$Ma = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix} = b$$

Method 2: keep the point fixed and move the coordinate system:
make one unit along the horizontal axis half as long as before;
make one unit along the vertical axis one-third as long as before.

Suppo

Move the Point vs Move the Coordinate Frame

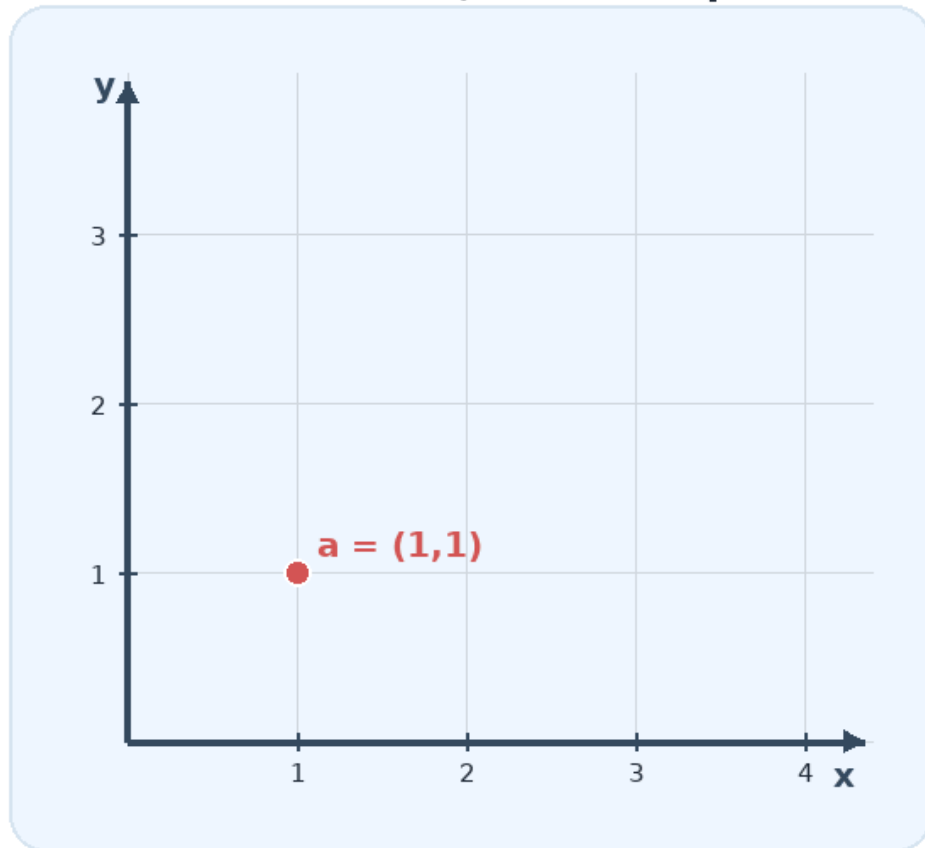
Two different stories, the same final coordinates (2,3).

Metho

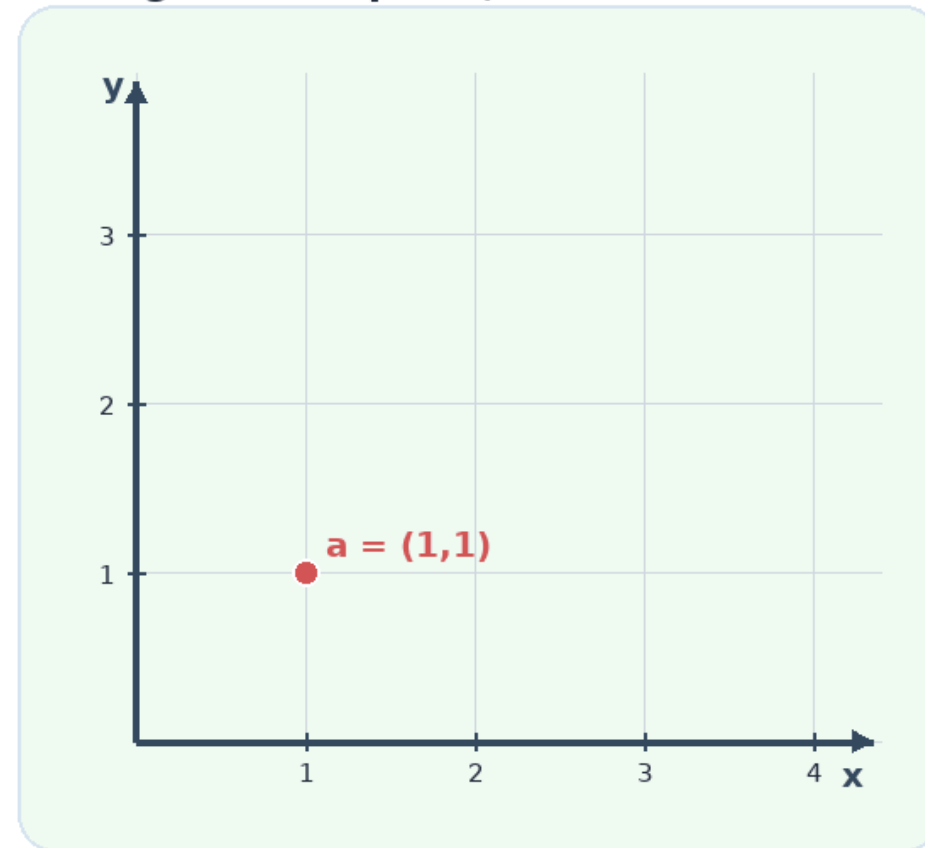
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$$\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$$

Left: fixed axes, move the point



Right: fixed point, shrink the units



Mathematics Refresh: Linear Algebra

Suppose we want the coordinate numbers of a point to change from $a=(1,1)$ to $b=(2,3)$

$$\begin{bmatrix} 2 \\ 3 \end{bmatrix} = b = Ma = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Interpret M as a transformation: the coordinate system remains fixed. The matrix acts. The vector moves

$$Ib = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = b = Ma = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

There is one abstract geometric vector. Its coordinates under the coordination system formed by the columns of M are a . Its coordinates under the standard coordinate system are b .

Placing M in front of a is almost like attaching an environmental declaration to the vector.